

Yu Duan

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Education

Sep 2023 – Jun 2028
(Expected)

Massachusetts Institute of Technology

- Ph.D. in Electrical Engineering and Computer Science

Sep 2019 – Jun 2023

Tsinghua University, Institute for Interdisciplinary Information Sciences

- B.Eng. in Computer Science and Technology

Experience

May 2026 – Aug 2026

Adobe Inc. – Research Scientist Intern

- Post-trained a large-scale image foundation model for precise, instruction-based image editing, using supervised fine-tuning and reinforcement learning
- Developed an efficient method for continuous image editing (“sliders”) that gives fine-grained control over edit strength
- *First-author submission to CVPR 2027 in preparation (continuing as an academic collaboration).*

Feb 2022 – Jul 2022

Massachusetts Institute of Technology, Department of Brain and Cognitive Sciences
– Visiting Student

- Built multi-system RNN models of human visual working memory (see Research Experience)

Technical Skills

- **Languages:** Python, C/C++
- **Frameworks:** PyTorch, JAX, Scikit-learn
- **Machine Learning:** Foundation Models, Diffusion and Flow Matching, Post-training (SFT / RL), Transformers, Time-series Forecasting, Multi-modal Generative Modeling
- **Neuroscience:** Neural Population Dynamics, Recurrent Network Modeling, Large-scale Neural Data Analysis

Selected Research Experience

May 2025 – Present	Causal and Generative Foundation Models for Neural Dynamics • Building foundation models that predict neural responses to optogenetic perturbations, moving from observational forecasting to causal (interventional) prediction • Developing flow-matching generative models that capture the full distribution of trajectories in noisy, high-dimensional dynamical systems
Feb 2024 – May 2025	POCO: Foundation Model for Neural Population Forecasting • Designed a population-conditioned forecasting architecture that trains a single model on neural population recordings across sessions, individuals and species • Established a large-scale multi-species benchmark, outperforming existing forecasting baselines and adapting rapidly to held-out sessions • <i>Published at NeurIPS 2025 (First Author).</i>
Mar 2021 – Sep 2022	Hebbian and Gradient-based Plasticity for Rapid Learning in RNNs • Developed RNNs that combine gradient-based training with local Hebbian plasticity rules, enabling rapid few-shot learning and robust memory • <i>Published at ICLR 2023 (First Author).</i>
Feb 2022 – Dec 2023	Sensory-Cognitive Models of Visual Working Memory • Showed that human-like working memory capacity limits emerge from architectural bottlenecks in interacting sensory and cognitive networks (co-first author; COSYNE 2023, CCN 2022)
Mar 2023 – Jun 2023	Vision-Language Representations for Neural Encoding • Aligned pretrained vision-language model representations (CLIP, ViT) with human fMRI voxel responses to natural images

Publications

2025	POCO: Scalable Neural Forecasting through Population Conditioning • Yu Duan, Hamza Tahir Chaudhry, Misha B. Ahrens, Christopher D Harvey, Matthew G Perich, Karl Deisseroth*, Kanaka Rajan* • NeurIPS 2025
2023	Hebbian and Gradient-based Plasticity Enables Robust Memory and Rapid Learning in RNNs • Yu Duan, Zhongfan Jia, Qian Li, Yi Zhong, Kaisheng Ma • ICLR 2023
2023	Natural constraints explain working memory capacity limitations in sensory-cognitive models • Yudi Xie*, Yu Duan*, Aohua Cheng, Pengcen Jiang, Christopher Cueva, Guangyu Robert Yang (*equal contribution) • Computational and Systems Neuroscience (COSYNE 2023) • Conference on Cognitive Computational Neuroscience (CCN 2022)

Honors

2022	Scholarship for Scientific Innovation, Tsinghua University
2021	Scholarship for Academic Excellence, Tsinghua University
2019	Freshman Scholarship, Tsinghua University
2017	Gold Medal, National Olympiad in Informatics, China